

COS-AT1

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO5: Design Pushdown Automata and analyze their equivalence with Context-Free Grammars through formal construction and proof techniques. (BL :3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

Constraint Based Problem

Code of Conduct:

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I 192372044 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Shaik Hafeeza Taranum

SIMATS ENGINEERING ASSESSMENT TOOLS

1. Design a Pushdown Automaton that accepts the language

$$L = \{ a^n b^n \mid n \geq 1 \}$$

Constraint: You are allowed to use only one stack symbol (besides initial symbol)

2. CFG to PDA Conversion with Limited Productions

Given a Context-Free Grammar:

$$S \rightarrow aSb \mid ab$$

Construct an equivalent PDA.

Constraint: PDA must be constructed using only 2 states

3. PDA to CFG Conversion with Restricted Variables

Convert a given PDA into an equivalent CFG.

Constraint: The resulting CFG must use at most 3 variables (non-terminals)

4. Turing Machine with Limited Tape Movement

Design a Turing Machine to recognize the language

$$L = \{ ww \mid w \in \{0,1\}^* \}$$

Constraint: The tape head can move only to the right (no left movement allowed)

5. Consider a language $L = \{ a^n b^n c^n \mid n \geq 1 \}$

Constraint: Design a Turing Machine for L

• PDA for $L = \{a^n b^m \mid n \geq 1\}$

Constraint

only one stack symbol decides the initial stack symbol is allowed
Let the stack symbols be

$$\Gamma = \{Z_0, A\}$$

• Z_0 = initial stack symbol

• A = The only Additional stack symbol

Idea

- For Every a , push one A onto the stack
- For Every b , pop one A
- Accept When all A 's been removed and only Z_0 remains.

$$M = \{Q, \Sigma, \Gamma, \delta, q_0, Z_0, F\}$$

where:

$$Q = \{q_0, q_1, q_f\}$$

$$\Sigma = \{a, b\}$$

$$\Gamma = \{Z_0, A\}$$

$$F = \{q_f\}$$

Transition Function

cs	input	stacktop	operation	Nextstate
q_0	a	Z_0	pushA	q_0
q_0	a	A	pushA	q_0
q_0	b	A	popA	q_1
q_1	b	A	popA	q_1
q_1	ϵ	Z_0	Accept	q_f

Formal Functions:

$$\delta(q_0, a, Z_0) = (q_0, AZ_0)$$

$$\delta(q_0, a, A) = (q_0, AA)$$

$$\delta(q_0, b, A) = (q_1, \epsilon)$$

$$\delta(q_1, b, A) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, Z_0) = (q_f, Z_0)$$

Conclusion: The PDA pushes one A for every a and pops one A for every 'b'. Therefore, it accepts exactly.

$$L = \{a^n b^n \mid n \geq 1\}$$

using only one additional stack symbol A

(2) CFG to PDA Conversion

Given CFG $S \rightarrow asb|ab$.

Constraint: The PDA must use only 2 states.

PDA Construction:

Let

$$Q = \{q_0, q_1\}$$

• q_0 : placing state

• q_1 : accepting state

Stack alphabet:

$$\Gamma = \{z_0, s, a, b\}$$

Input alphabet:

$$\Sigma = \{a, b\}$$

Initial stack symbol: z_0

Step 1: push the stack variable:

$$\delta(q_0, \epsilon, z_0) = (q_0, S, z_0)$$

Step 2: Replace S using grammar productions.

For: $S \rightarrow asb$

$$\delta(q_0, \epsilon, S) = (q_0, bSa)$$

$S \rightarrow ab$

$$\delta(q_0, \epsilon, S) = (q_0, ba)$$

Step 3: Math Transitions:

$$\delta(q_0, a, a) = (q_0, \epsilon)$$

$$\delta(q_0, b, b) = (q_0, \epsilon)$$

Step 4: Accept:

$$\delta(q_0, \epsilon, z_0) = (q_1, z_0)$$

Conclusion: The PDA uses only two states and simulate production $S \rightarrow asb|ab$

Thus, PDA accepts string such as: $ab, aabb, aaabbb, \dots$

PDA to CFG conversion with at most 3 variables.

General Approach:

Suppose PDA has states q_0 and q_1

- Reading a 's and pushing a stack symbol.
- Reading b 's and popping the stack symbol.
- Accepting when stack becomes empty.

$$L = \{a^n b^m | n \geq m\}$$

$$\text{CFG} = S \rightarrow aSb | ab$$

$$\text{variables } V = \{S\}$$

only one terminal is actually required, which satisfies the constraint of at most 3 variables.

Explanation:

$$S \rightarrow aSb$$

$$S \rightarrow ab$$

$$S \rightarrow aSb$$

$$\Rightarrow aasbb$$

$$\Rightarrow aaabb$$

Thus equivalent CFG is

$$\boxed{S \rightarrow aSb | ab}$$

which uses only one terminal.

4) Turing machine for $L = \{ww | w \in \{0,1\}^*\}$

constraint: The tape head can move only to the right.

Such a Turing machine cannot recognise $L = \{ww\}$

Reason: A standard one-tape Turing machine that can move only to the right cannot move back to:

- compare symbols in the first half with corresponding symbols in second half
- mark previously checked symbols.
- return to an earlier portion.

For ex:

$w = 01$

$ww = 0101$

- first 0 with third 0,
- second 1, with fourth 1.

⇒ This requires remembering an arbitrary amount of information about the first half (or moving back on tape)

⇒ A machine restricted only right movement behaves like finite automation, because once it moves past a symbol, it cannot inspect it again

$$L = \{ ww \mid w \in \{0,1\}^* \}$$

is not a regular language

No standard one-tape right-only Turing machine can recognize $\{ ww \}$

Final Answer:

The given constraint makes the problem impossible. A Turing machine must be allowed to move both left and right to recognize $L = \{ ww \}$.

5) Turing machine for $L = \{ a^n b^m c^n \mid n \geq 1 \}$

Idea:

The Turing machine repeatedly

1. Find the leftmost unmarked a , marks it x .
2. moves right and find the first unmarked b , marking it y .
3. Moves right and finds first unmarked c , marking it z .
4. Returns to beginning
5. Repeat until all symbols are marked
6. Accepts if all a, b and c symbols are matched

Tape symbols: $\Sigma = \{a, b, c, x, y, z, B\}$

where:

- x marks a processed 'a'
- y marks a processed 'b'
- z marks a processed 'c'
- B (marks a processed) blank

working procedure:

$a \rightarrow x$

$b \rightarrow y$

$c \rightarrow z$

Ex: input = aaabbbccc
xaabbbccc
xaaxbbccc
xaaaybbzccc

eventually xxxxyyyzzz

conclusion:

The Turing machine accepts?

$L = \{a^n b^m c^n \mid n \geq 1\}$

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

86/100

